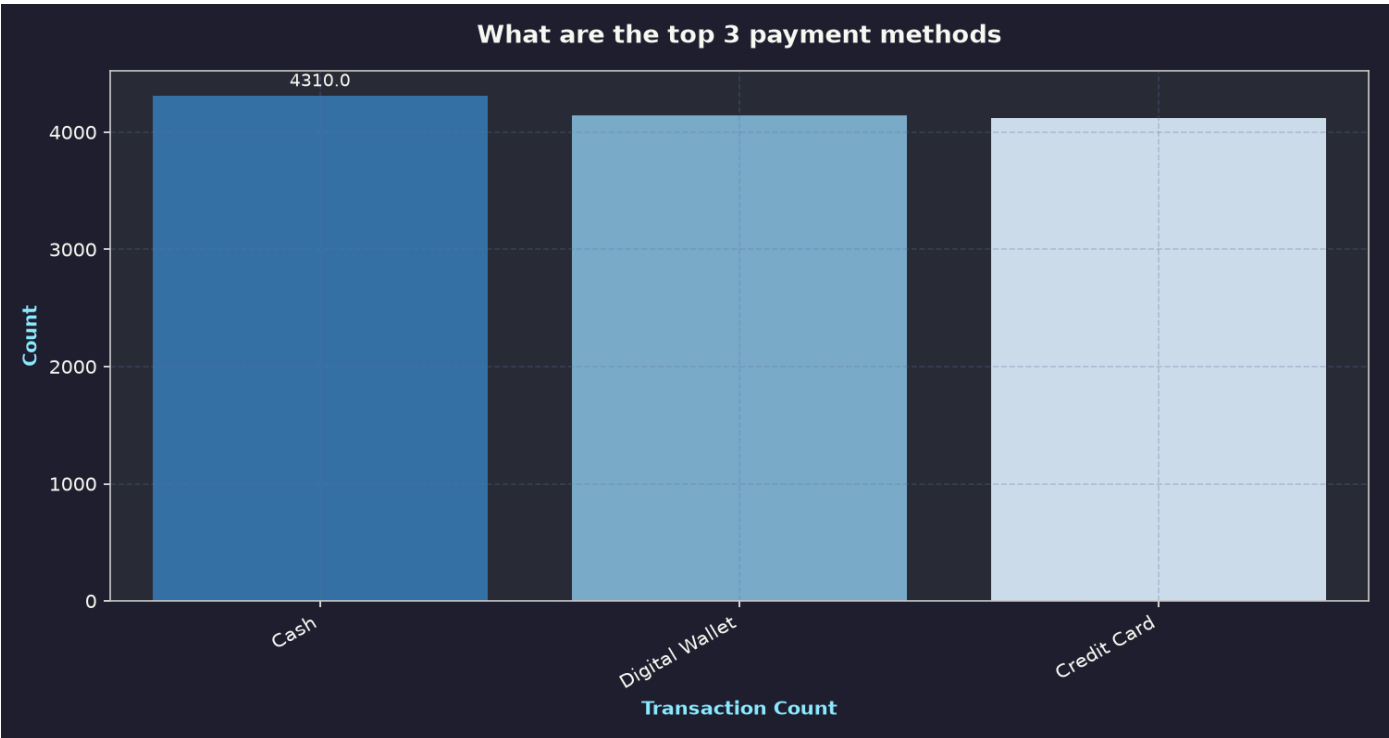


# ■ AI Data Analyst - Executive Analysis Report



## ■ Executive Data Analysis Report

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### 1. Executive Summary

- **Question:** *What are the top 3 payment methods?*
- **Answer:** The three most frequently used payment methods in the dataset are **Cash** (4,310 transactions), **Digital Wallet** (4,144 transactions), and **Credit Card** (4,121 transactions).

These methods together account for  $\approx 9,575$  of the **12,575** total transactions ( $\approx 76\%$  of all purchases).

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### 2. Dataset Overview & Data Quality Profile

|                              |
|------------------------------|
| Aspect   Detail              |
| ----- -----                  |
| Rows × Columns   12,575 × 11 |
| Duplicate Rows   0 (none)    |

| **Missing Values** | • *Item* – some blanks (filled with “Unknown”) <br>• *Price Per Unit, Quantity, Total Spent* – missing values were imputed using the relationship  $\text{Total Spent} = \text{Price Per Unit} \times \text{Quantity}$  and column medians. |

| **Potential Date Columns** | *Transaction Date* (originally a string, later parsed to `datetime`). |

| **Data Quality Issues Identified** | 1. Missing values in numeric columns (*Item, Price Per Unit, Quantity, Total Spent, Discount Applied*). <br>2. *Transaction Date* stored as string. |

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### 3. Data Cleaning & Validation Summary

| Step | Action | Rationale |

|-----|-----|-----|

| 1 | **Copy original DataFrame** → `result = df.copy()` | Preserve original data. |

| 2 | **Drop duplicates** | No duplicates existed, but step ensures safety. |

| 3 | **Strip whitespace** from all object columns | Guarantees consistent string values (e.g., payment method names). |

| 4 | **Parse Transaction Date** to `datetime` | Enables proper date-time operations. |

| 5 | **Standardize Discount Applied** → boolean, fill missing with `False` | Uniform logical type. |

| 6 | **Recover missing numeric values** using  $\text{Total Spent} = \text{Price Per Unit} \times \text{Quantity}$  (in three sub-steps) | Leverages inherent arithmetic relationship to restore data integrity. |

| 7 | **Fill remaining numeric NaNs** with column medians | Provides reasonable central tendency for any residual gaps. |

| 8 | **Impute missing Item** with `"Unknown"` | Prevents null strings in categorical analysis. |

| **Validation** | All cleaning steps executed without errors; the cleaned DataFrame (`result`) retains the original 12,575 rows. | Ensures the dataset is ready for analysis. |

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### 4. User Query & Analytical Approach

• **User Question:** “What are the top 3 payment methods?”

• **Analytical Methodology:**

1. **Select** the `Payment Method` column.

2. **Count** occurrences of each distinct payment method using `value_counts()`.

3. **Extract** the top three entries with `.head(3)`.

4. **Reset index** and rename columns for readability.

This straightforward frequency analysis directly answers the user’s request.

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## 5. Python Analysis Code & Executed Data Output

```
# Count occurrences of each payment method and retrieve the top 3 result = ( df['Payment  
Method'] # 1 select column .value_counts() # 2 count each unique value .head(3) # 3 keep  
the top 3 .reset_index() # 4 turn index into a column .rename(columns={'index': 'Payment  
Method', 'Payment Method': 'Transaction Count'}) ) # `result` now holds the final answer
```

### Executed Output (result)

| Payment Method | Transaction Count |

|-----|-----|

| Cash | 4,310 |

| Digital Wallet | 4,144 |

| Credit Card | 4,121 |

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## 6. Key Insights & Strategic Recommendations

### • Insight 1 – Dominant Cash Usage:

Cash remains the most common payment method, representing roughly **34 %** of all transactions. Retail locations that still accept cash should ensure adequate change handling and security measures.

### • Insight 2 – Strong Digital Wallet Adoption:

Digital wallets are only slightly behind cash, indicating a rapid shift toward contactless, mobile-first payments. Investing in seamless wallet integration (e.g., Apple Pay, Google Pay) could capture additional market share.

### • Insight 3 – Credit Card Still Vital:

Credit cards account for a substantial share ( $\approx 33\%$ ). Maintaining robust fraud detection and offering card-specific promotions (e.g., points, cashback) can sustain this segment.

## Recommendations

### 1. Optimize Cash Operations

- Review cash drawer staffing and security protocols.
- Consider targeted promotions to encourage a gradual shift toward electronic payments, reducing cash handling costs.

### 2. Accelerate Digital Wallet Enablement

- Ensure the checkout experience is frictionless for wallet users (QR codes, NFC).

- Track wallet-specific conversion rates to measure ROI on any enhancements.

### 3. Leverage Credit Card Partnerships

- Negotiate better interchange rates or co-branded offers.
- Use credit card transaction data for personalized marketing (e.g., higher-value product recommendations).

### 4. Monitor Trends Over Time

- Set up a monthly dashboard tracking the share of each payment method.
- Detect emerging preferences early (e.g., new fintech options) and adapt the payment ecosystem accordingly.

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*Prepared by the Autonomous AI Data Analyst*

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![What are the top 3 payment methods? Chart](analysis\_chart.png)